PARKING SENSOR SYSTEM

id092200024200

Outline

- The parking sensor system detects obstructions in the blind spot (vehicle rear/corners) to a wide extent using ultrasonic sensors and notifies the driver of the obstructions using the indicator and buzzer.

Warning

- **The parking sensor system is a device to assist the driver in confirming safety. It is not a substitute for the driver and cannot guarantee 100 % safety. Do not rely completely on the parking sensor system and drive while confirming the safety of the surrounding area visually.**

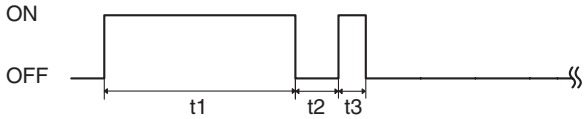
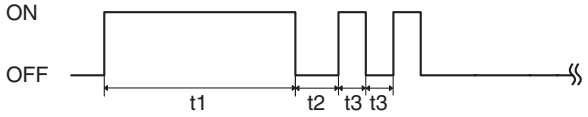
- The following circumstances may obstruct the ultrasonic sensor resulting in an accident:
  - — Ice, snow, or dirt is adhering to ultrasonic sensor
  - — Ultrasonic sensor is covered by foreign matter (such as bumper stickers)
  - — Shock is applied to bumper in accident and ultrasonic sensor installation angle is deviated
  - — Ultrasonic sensor is exposed to heavy rain or water-spray from road
  - — There is another device emitting ultrasonic waves (such as ultrasonic sensor of another vehicle) near ultrasonic sensor
- The parking sensor system may not operate normally under the following conditions:
  - — Ice, snow, or dirt is adhering to ultrasonic sensor
  - — Ultrasonic sensor is frozen
  - — Vehicle is in tilted position
  - — Vehicle is parked on steep incline or there is difference in height between lines
  - — Obstructions have approached too close to ultrasonic sensor
- The ultrasonic sensor may not detect the following items:
  - — Thin objects such as wire or rope
  - — Items which absorb ultrasonic waves easily, such as cotton or snow
  - — Angular-shaped objects
  - — Tall objects protruding at the top
  - — Short objects
  - — Objects located immediately below bumper, such as wheel stoppers
- The warning alert may activate while driving near parked vehicles/guardrails/walls on a narrow road, or while driving on a slope, gravel road, or grass field. This is a normal effect caused by the system's response to ultrasonic waves, and does not indicate a malfunction.

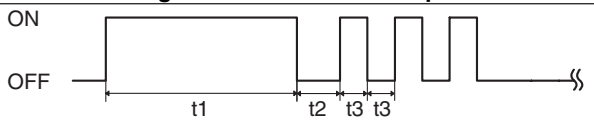
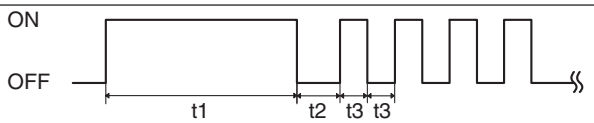
Function

- The ultrasonic sensor located on the rear bumper detects an obstruction and the parking sensor control module informs the driver of the distance between the vehicle and the obstruction using the parking sensor buzzer.

Ultrasonic sensor malfunction detection function

- If the parking sensor control module detects that there is a malfunction in the ultrasonic sensor circuit, the sound is activated in the following pattern.

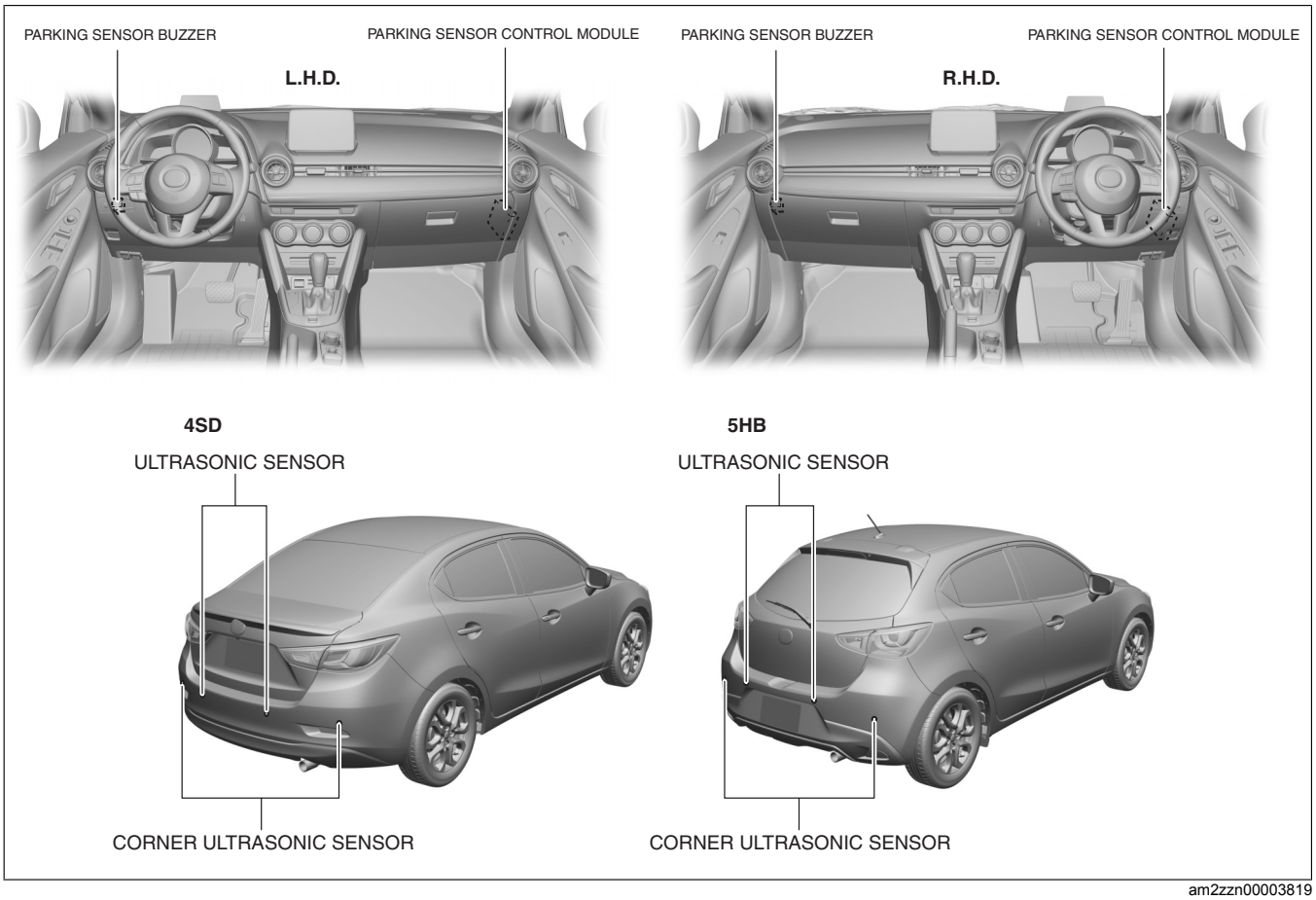
Malfunction detection location	Parking sensor buzzer sound pattern
Corner ultrasonic sensor (LH)	 t1: APPROX. 1.8—2.2 s t2: APPROX. 0.45—0.55 s t3: APPROX. 0.27—0.33 s
Ultrasonic sensor (LH)	 t1: APPROX. 1.8—2.2 s t2: APPROX. 0.45—0.55 s t3: APPROX. 0.27—0.33 s

Malfunction detection location	Parking sensor buzzer sound pattern
Ultrasonic sensor (RH)	 t1: APPROX. 1.8—2.2 s t2: APPROX. 0.45—0.55 s t3: APPROX. 0.27—0.33 s
Corner ultrasonic sensor (RH)	 t1: APPROX. 1.8—2.2 s t2: APPROX. 0.45—0.55 s t3: APPROX. 0.27—0.33 s

Volume adjustment function

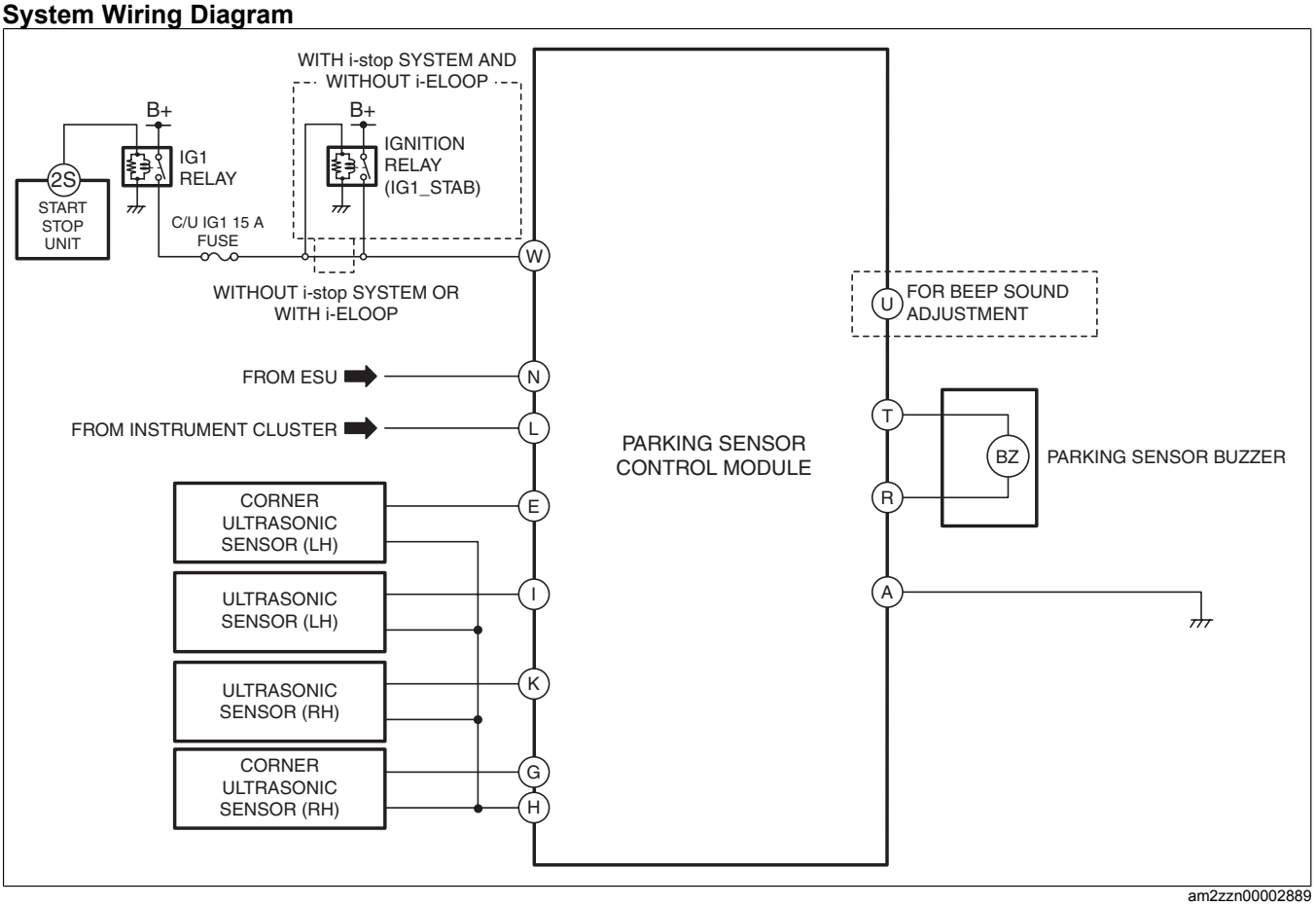
- The volume of the parking sensor buzzer can be adjusted. Refer to the Workshop Manual for the adjustment procedure.

Structural View



am2zzn00003819

System Wiring Diagram



am2zzn00002889

Operation

Operation condition

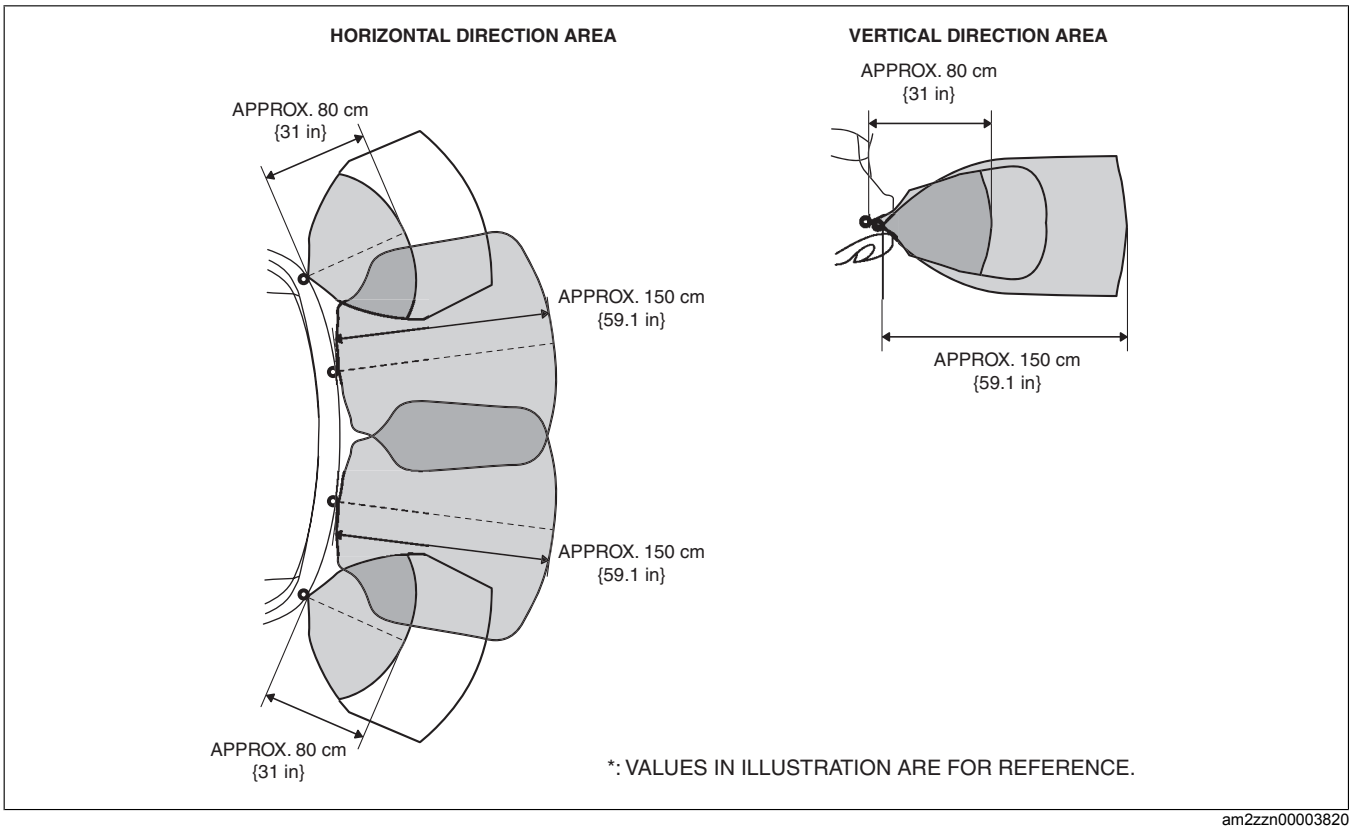
- When all of the following conditions are met, the parking sensor system operates and the parking sensor buzzer activates for **approx. 0.8 s**.
  - — Ignition switched ON (engine on)
  - — R position (ATX)/reverse gear (MTX)
- If the parking sensor system operates, the ultrasonic sensor starts to detect obstructions.For the ultrasonic sensor which is operating, refer to the following table.

×: Operates, —: Does not operate

Vehicle condition		Ultrasonic sensor	
Selector lever (ATX) Shift lever (MTX)	**Vehicle speed**	**Corner ultrasonic sensor**	**Ultrasonic sensor**
R position (ATX)/reverse gear (MTX)	**Less than 10 km/h {6.2 mph}**	×	×
	10 km/h {6.2 mph} or more	—	×

Obstruction Detection Area

- Each ultrasonic sensor detects obstructions in the following areas.

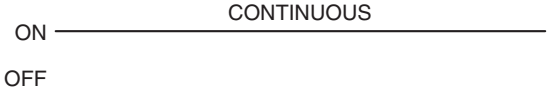
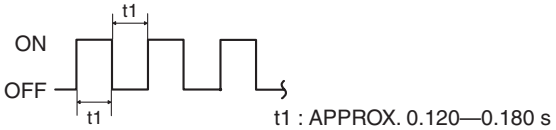


am2zzn00003820

Parking sensor buzzer sound pattern
Obstructions detected

- The changes in the amount of time the parking sensor buzzer activates are as follows:

Ultrasonic sensor	Zone	Distance from obstruction	Parking sensor buzzer sound pattern
Corner ultrasonic sensor	Short distance	**Approx. 0.0—25 cm {0.0—9.8 in}**	CONTINUOUS ON OFF
	Intermediate distance	**Approx. 25—38 cm {9.9—14.0 in}**	t1 ON OFF t1 t1 : APPROX. 0.055—0.095 s
	Long distance	**Approx. 38—55 cm {15—21 in}**	t1 ON OFF t1 t1 : APPROX. 0.120—0.180 s

Ultrasonic sensor	Zone	Distance from obstruction	Parking sensor buzzer sound pattern
Ultrasonic sensor	Short distance	**Approx. 0—35 cm {0—13 in}**	
	Intermediate distance	**Approx. 35—45 cm {14—17 in}**	
	Long distance	**Approx. 45—60 cm {18—23 in}**	
	Maximum long distance	**Approx. 60—150 cm {24.0—59.0 in}**	